Exp 7 MATLAB CODE

```
clc;
clear;
close all;

Fs = 100000;
t = 0:1/Fs:0.01;

Am = 1;
fm = 1000;
fc = 10000;
kf = 5000;

m = Am*cos(2*pi*fm*t);

int_m = cumsum(m)/Fs;
s_fm = cos(2*pi*fc*t + 2*pi*kf*int_m);

frequency_deviation = kf * Am;
modulation_index = frequency_deviation / fm;

fprintf('Message Frequency = %d Hz\n', fm);
fprintf('Frequency Deviation = %.2f Hz\n', frequency_deviation);
fprintf('FM Modulation Index = %.2f\n', modulation_index);

figure;

subplot(3,1,1);
```

```
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,1,2);
plot(t,s_fm);
title('Frequency Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,1,3);
N = length(s_fm);
f = (-N/2:N/2-1)*(Fs/N);
S = fftshift(abs(fft(s_fm))/N);
plot(f,S);
title('FM Signal Frequency Spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
xlim([0 20000]);
grid on;

% Verification
fprintf('\nVerification:\n');
fprintf('Modulation Index = Frequency Deviation / Message Frequency\n');
fprintf('Modulation Index = %.2f / %d = %.2f\n', ...
```

frequency_deviation, fm, modulation_index);

